

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari · Avijeeet Singh — Vortex AI · paper doi:10.5281/zenodo.20706893 (v0.1.0; current method: whitepaper v3, emem 213e273) · measurements 29–30 Sep 2026 · scripts: github.com/Vortex-AI/esa_poster

Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?

B receives an **address**, not a sentence. emem stores each observation as a signed record named by the **BLAKE3 hash of its own bytes**. B fetches the bytes, re-hashes them, verifies the signature offline, and can walk the record back to the source pixel in the Sentinel-2 file.

Token grammar: emem:fact:<cell64>:<fact_cid>, cell64 is a 64-bit ≈10 m grid cell; fact_cid = base32(BLAKE3-256(emem-CBOR bytes)). A token cited under the wrong cell returns HTTP 409.



2 · FULL TRACE

From one token to the source pixel, with no emem code

A 698-line script (Python stdlib, blake3, cbor2, pynacl) checked **15 links in 17.9 s, all verified**. Each link states the trust that remains after the check.

1–3	bytes → name	1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... · trust left: none
4–5	cell binding	cell inside the hashed body = cell in the token; cell decodes to the lat/lng the derivation used · none
6–7	derivation	scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · arg positions come from emem’s source, not a published schema
8–9	upstream	Planetary Computer STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum; the floor pixel rule is emem’s choice
10–11	receipt	ed25519 over the served request, cells and fact_cids; Merkle batch binding · proves what emem.dev served, not that it is true
12–14	log	attestation at leaf 2,457,078, included under the signed head 2,556,451, consistent with the head geo.qa co-signed · geo.qa is also run by Vortex AI
15	key → domain	the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

The whole proof packs into a 4,906-byte bundle that re-verifies with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it. research/repro/v8/trace_fact.py · verify_bundle.py · §22

4 · ONE TOKEN, MANY CLIENTS

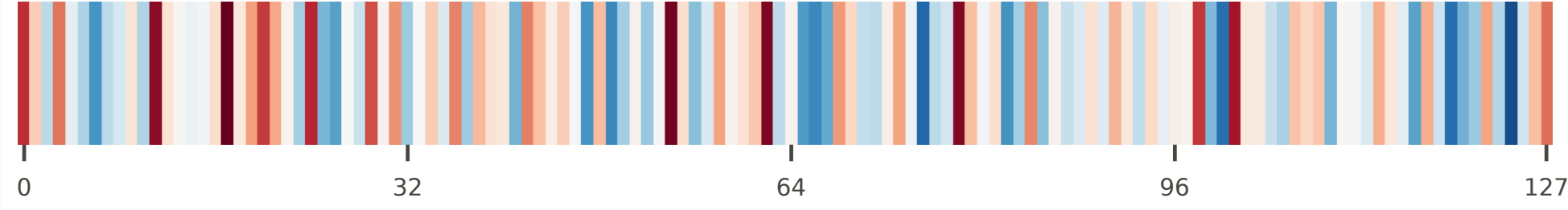
Ten client paths, one record

PATH	CID	RE-HASH	RECEIPT
REST · MCP (Streamable HTTP) · A2A JSON-RPC	oj5cecci	3/3	3/3
Python SDK · TypeScript SDK	oj5cecci	2/2	2/2
official MCP SDKs, LangChain MCP adapters, LlamaIndex tool	oj5cecci	4/4	3/3 †
stock blake3 + cbor2 on the bytes	oj5cecci	1/1	—
same bytes from an untrusted mirror; tampered copy	j47f4xb3	rejected	

3 repetitions each, 30 Sep 2026; value 0.4708994708994709 on every path. † the LlamaIndex tool drops the receipt. One responder (emem.dev) behind all paths; this shows client and protocol independence, not independent servers.

7 · OVER FOUNDATION-MODEL EMBEDDINGS

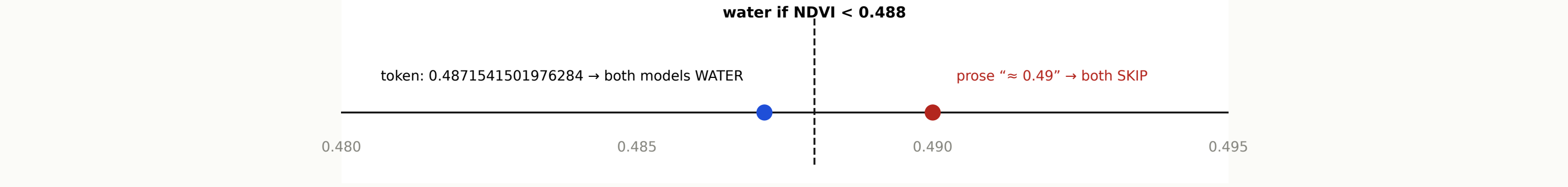
A Tessera vector, re-derived by anyone



At the same cell, emem signs the 128-D **TESSERA v1** (Cambridge; Sentinel-1 + Sentinel-2, 2024) embedding as a model_output record §9. Two range reads of the public product on Source Cooperative (128 B int8 + 4 B scale) give all 128 floats back with max |Δ| = 0; 4 of 4 facts tested match. emem re-serves this product; it does not run the encoder, and the record says so (tamper_evidence: attester_only). Encoders emem did run, such as Clay v1.5 §10, pin the checkpoint by BLAKE2b.

10 · WHY A PARAPHRASE IS NOT A CITATION

Two models agreed on a rounded number and made the same wrong call



PRE-REGISTERED, GEMMA-4-12B + QWEN2.5-7B, ONE HOST	CORRECT	AGREE
full context (control)	72/72	36/36
after a shared compaction summary	0/72	3/36

agreement exceeds accuracy under compaction: Fisher one-sided p = 0.035 §17; handoff n = 20 per arm; prose 2, dense retrieval 8, BM25 20, bundle 20 §18

The trap value 0.4871541501976284 §21 is the value as signed on 17 Jul. It is a pre-fix read of the pixel south (section 3); the experiment tests fidelity to the cited value, not the ground. BM25 ties the bundle on short handoffs. Related work: telephone-game drift (Perez et al., ICLR 2025), “Lost in Compaction” (arXiv 2608.11242), pointer-based compaction ARC (arXiv 2607.25066).

12 · EMEMIFY YOUR WORK

Five steps, each run on 29–30 Sep

- 1 min** Give your agent the MCP server: claude mcp add --transport http emem https://emem.dev/mcp (18 core tools, 114 in all).
- 10 min** Cite a value in a paper by its token and receipt; the research-grade-citation skill writes the methods sentence.
- 1 hour** Register your own result, signed by your key: an NDVI change of −0.1298 (25 Sep 2025 → 2026) at this cell §11; emem recomputed it, ulp_gap 0.
- 1 day** Tokenise your files; they stay where they are; a signed pointer note holds their chunk hashes. This board is one §1.
- 1 week** Run your own node: docker run ghcr.io/vortexai-ai/emem. A local node re-read this pixel: bit-identical NDVI, a different fact_cid (its own key signs).

Apache-2.0. pip install ememdev · npm i @vortexai/emem · A2A card at emem.dev/well-known/agent-card.json (signed, EdDSA).

5 · TWO LLMs, EMEM AS THE ONLY PLUGIN

Pre-registered handoff: the token fails closed

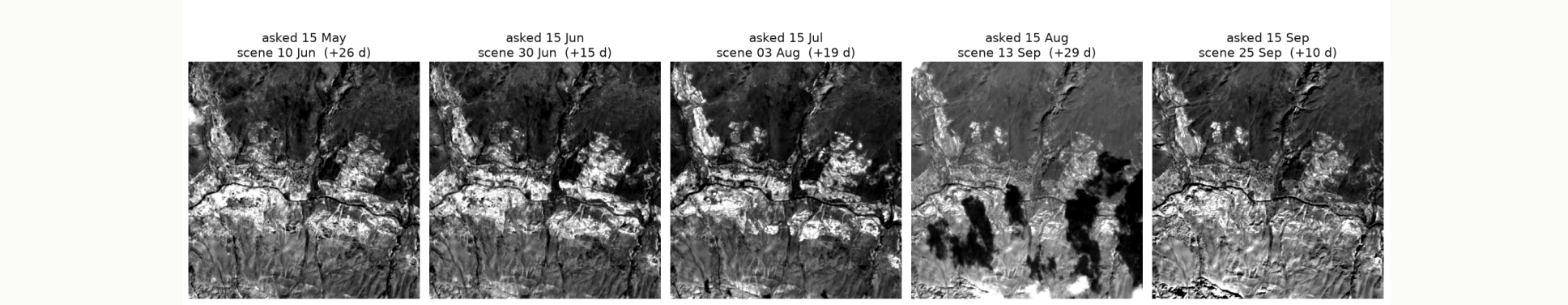
B RECEIVES FROM A	B CORRECT	B CHECKED
A’s token	10/10	resolve, re-hash, receipt 10/10
A’s own prose (16 digits)	10/10	nothing to check
prose rounded to “0.47”	0/5 —	
token with a forged cell	5/5 declined	409 caught

A: Claude Sonnet 5.5 with emem as its only MCP server returned the exact token in 10/10 runs (13 s, 3.3 tool calls each). B: Claude Haiku 4.5. Rule: IRRIGATE if NDVI ≤ 0.4705, a **constructed** threshold that 2-decimal rounding crosses. Token vs rounded prose: Fisher p = 0.0003; token vs full prose: a tie.

Pre-registration hashed before trial 1 (research/repro/data/v8/prereg.md). One model family for both agents; the local Qwen arm did not finish.

8 · FIELDS AND FIELDS THROUGH TIME

One cube, five scenes, each date gap stated



An emem:cube: token §6 is a signed manifest of five raster records; each member states the date asked for and the scene used (+10 to +29 days). The 13 Sep member is cloudy: scenes are chosen by scene-level cloud cover (≤ 40 %, relaxed in tiers), not a pixel mask.

11 · INTEGRITY ELSEWHERE, FOR THE SAME SCENE

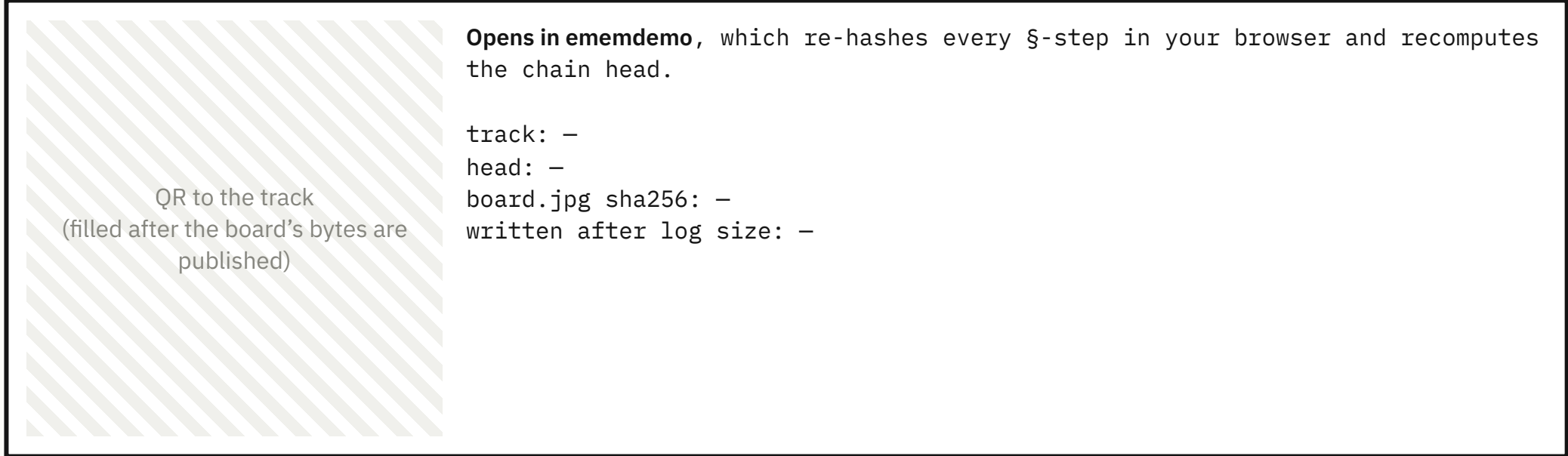
What an agent can check today, by source

WHAT THE AGENT IS HANDED	INTEGRITY IT CARRIES
Planetary Computer: 281,898,500 B B08 COG URL	opaque ETag, no checksum
Copernicus Data Space: 1,187,971,637 B SAFE	MD5 + BLAKE3 of the whole product, unsigned
NASA Earthdata MCP: a granule concept id	catalogue revision; older revisions return 404
MCP tool results in general (spec 2026-07-28)	none; proposed in issue #3354
emem: one value	BLAKE3 name of the 1,115-byte record, ed25519 receipt, log inclusion

These layers compose: STAC and COG find and read data, openEO and Earth Engine compute, retrieval ranks, emem names the exact value an agent cited so another agent can check it. emem does not yet store a hash of the upstream bytes (Source .hash is empty), so the upstream read is checked by re-reading the pixel, as in section 2.

14 · SEAL

Check this board



A file cannot contain its own hash, so this box is the only part of the board not inside the published bytes §3. Signed by the poster key njedkgt...; scripts to re-check everything without the viewer: research/repro/v8/.